Forum:	Economic and Social Council (ECOSOC)
Issue:	Addressing the issue of Balancing economic feasibility and environmental sustainability in the adoption of next-generation nuclear energy technologies
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Position:	President

Personal Introduction

Dear delegates and all other readers,

My name is Konstantinos Grafopoulos and I have the utmost honour and responsibility of serving as this year's conference President of the Economic and Social Council (ECOSOC) committee. I am an 11th-grade student at George Zois School and the Model United Nations conference hosted by Moraitis School has a special place in my heart and so, I hope to create a warm and welcoming atmosphere for all of you, along with my two Deputy Presidents.

The topics of this year's ECOSOC are centred on the economic and social implications of illegal oil transfers, water scarcity prevention mechanisms and the means of achieving a balanced approach to the utilisation of nuclear energy. Specifically, in this study guide, the topic at hand is: "Balancing economic feasibility and environmental sustainability in the adoption of next-generation nuclear energy technologies". Balancing nuclear development with environmental sustainability presents a significant challenge.

Regarding any questions on the topic of this agenda or any assistance needed for your preparation, please do not hesitate to contact me via my email address: grafopoulos.k@zois.edu.gr . grafopoulos.k@zois.edu.grgrafopoulos.k@zois.edu.gr

In conclusion, I hope this study guide helps you understand the different aspects of the topic, however, I encourage you to extend your research to acquire a more comprehensive understanding.

See you at the conference!

Kind regards,

Konstantinos Grafopoulos.

Introduction

The global demand for clean and reliable energy continues to grow and consequently, governments and related industries face the challenge of reducing greenhouse gas emissions while ensuring energy security and economic development. Climate change, population growth and increasing electricity consumption are among the factors that have intensified the search for low-carbon energy sources which are able to provide consistent power generation. Out of all the available options, next-generation nuclear energy technologies have stood out as a promising solution owing to their ability to generate large amounts of electricity with very low direct carbon emissions. Advanced nuclear systems such as Small Modular Reactors (SMRs), Generation IV reactors and advanced fuel cycle technologies, seem to offer an alternative that combines potential improvements in safety, efficiency, waste management and operational flexibility compared to conventional nuclear power procedures. Still, although their technological potential is undeniable, the adoption of these systems is not yet widespread and depends highly on achieving an effective balance between economic feasibility, environmental sustainability and safety guarantees.

Economic feasibility remains one of the most significant barriers to deploying next-generation nuclear technologies. Although advanced reactors may offer lower operating costs, improved efficiency and broader service lifetimes, they often require substantial upfront capital investment, extensive regulatory approval and long development timelines which can make projects financially unviable. Financing large-scale nuclear projects proves to be challenging due to uncertain returns on investment, market conditions and competition from lower-cost renewable energy sources such as solar and wind power. Furthermore, the costs associated with research and construction, workforce training and infrastructure development may discourage both public and private investors. Thus, policymakers and industry leaders must take into consideration whether the long-term economic benefits, including reliable energy, potentially lower fuel costs and stable electricity prices, are sufficient to justify the initial financial commitments.

At the same time, environmental sustainability is fueling interest in the next-generation nuclear technologies. Unlike fossil-fuel-based power generators, nuclear energy produces very low greenhouse gas emissions when operated, making it a crucial tool for achieving global climate goals and reducing reliance on carbon-intensive energy sources. Many reactor designs also aim to improve fuel efficiency, reduce the volume and radiotoxicity of radioactive waste, whilst also enhancing safety systems that hope to minimize the risk of severe accidents. Some advanced reactor systems may be designed to use recycled nuclear fuel, further cutting back on waste and improving resource utilization. Nevertheless, environmental concerns remain a pivotal issue, be it radioactive waste disposal, uranium mining, water consumption, ecosystem impacts and the long-term management of decommissioned nuclear facilities.

Key Terms and Definitions

Small Modular Reactors (SMRs)

Small Modular Reactors (SMRs) are advanced nuclear reactors designed to generate electricity on a smaller scale than conventional nuclear power plants, typically producing up to 300 megawatts of electrical power per module. Unlike traditional large-scale nuclear reactors, SMRs are designed to allow major components to be manufactured in factories and transported to installation sites.

Generation IV Nuclear Reactors

Generation IV nuclear reactors represent a group of advanced nuclear technologies designed to improve upon earlier generations of nuclear power systems.

Environmental Sustainability

Environmental sustainability refers to meeting present energy demands while protecting natural resources and minimizing environmental damage for future generations.

Economic Feasibility

Economic feasibility refers to whether a project can be successfully developed, operated, and maintained at a financially sustainable cost.

Nuclear-energy generators

Nuclear-energy generators are electricity-producing systems that use energy released through controlled nuclear fission. During nuclear fission, the nucleus of a heavy atom, usually uranium, splits into smaller nuclei, releasing heat. This thermal energy is transferred through a cooling system and ultimately used to power turbines connected to electrical generators.

Fossil-fuel generators

Fossil-fuel generators produce electricity by burning coal, oil, or natural gas. The heat released through combustion is used to drive turbines or engines connected to electrical generators. These technologies have historically formed the foundation of industrial electricity production because of their established infrastructure and ability to provide controllable power.

Waste Management

Nuclear waste management refers to the safe handling, treatment, transportation, storage, and disposal of radioactive materials generated through nuclear activities.

Large-scale financing

Large-scale financing refers to obtaining substantial financial resources for infrastructure projects requiring major initial investments. Nuclear power plants are among the most capital-intensive energy projects because they involve sophisticated engineering, extensive safety systems, specialized personnel and lengthy construction periods.

Background Information

Historical Background

Fossil-Fuel-Based Technologies

For centuries, societies relied primarily on wood, water, wind and above all human or animal labor to meet their energy needs. The development of fossil-fuel technologies significantly transformed energy production by enabling the generation of larger quantities of mechanical power and electricity. Coal became a particularly important energy source because of its availability and high energy density. Later, petroleum and natural gas expanded the range of available energy technologies. Fossil fuels provided the foundation for industrial growth and urbanization. However, their widespread consumption also established energy systems dependent on finite resources and associated with significant environmental consequences.

The Industrial Revolution

The Industrial Revolution, beginning in Britain during the eighteenth century, marked a major turning point in global energy consumption. Coal-powered steam engines enabled the mechanization of manufacturing, transportation, and mining. Industrial production increased significantly, while the development of railways and steamships accelerated economic activity. During the nineteenth and twentieth centuries, electricity became increasingly important. Coal-fired power stations and, subsequently, oil- and natural-gas-powered facilities expanded electricity access. Although these developments improved living standards and productivity, they also led to a considerable increase in atmospheric pollution and greenhouse gas emissions.

Transition from Fossil Fuels to Alternative Forms of Generators

Growing concerns regarding environmental degradation, resource security, and climate change encouraged governments to explore alternatives to conventional fossil-fuel electricity generation. Hydroelectricity became one of the earliest commercially important alternatives, followed by the development of nuclear energy in the twentieth century. More recently, technological improvements and declining costs have supported the rapid expansion of solar and wind power. Nevertheless, replacing fossil fuels presents challenges involving energy storage, electricity-grid reliability, investment requirements, and regional differences in energy resources. Nuclear power has consequently been considered by some governments as a complementary source of low-carbon electricity capable of operating independently of weather conditions.

First Introduction of Nuclear-Energy Generators

Nuclear electricity generation emerged from scientific discoveries concerning radioactivity and nuclear fission during the late nineteenth and early twentieth centuries. In 1938, Otto Hahn and Fritz Strassmann conducted experiments that led to the discovery of nuclear fission, which was subsequently explained by Lise Meitner and Otto Frisch. The first controlled, self-sustaining nuclear chain reaction was achieved in Chicago in 1942 under the leadership of Enrico Fermi. Following the Second World War, attention increasingly turned toward the peaceful applications of nuclear

technology. In 1954, the Obninsk Nuclear Power Plant in the Soviet Union began supplying electricity to a power grid, marking an important milestone in civilian nuclear generation. In 1956, Calder Hall in the United Kingdom began commercial-scale nuclear electricity production. These developments demonstrated that nuclear energy could provide electricity on a large scale and initiated the expansion of civilian nuclear programs worldwide.

Economic Background

Economic Feasibility of Nuclear Energy Generation

The economic feasibility of nuclear power has remained a central issue since its introduction. Unlike fossil-fuel power plants, where fuel purchases can represent a substantial proportion of operating costs, nuclear facilities require particularly large upfront investments. Construction demands highly specialised materials, complex safety systems, and extensive regulatory assessment. These requirements can increase both costs and construction times. Nevertheless, once operational, nuclear power plants can produce electricity for several decades, with operating lifetimes often extended through modernization and safety improvements. Economic evaluations must consider the cost of capital, the availability of skilled workers, construction schedules, maintenance, nuclear fuel, waste management, and decommissioning. Nuclear electricity may be economically attractive under certain circumstances, particularly when governments provide stable financing and projects are completed efficiently; its competitiveness varies considerably between countries and individual projects.

Earlier Economic Investments in Nuclear Power Plants

During the 1950s and 1960s, several industrialized countries began investing heavily in civilian nuclear programs. Governments considered nuclear power a strategic technology that could strengthen energy independence, support industrial development, and meet growing electricity demand. France, the United States, the United Kingdom, Japan, and the Soviet Union became major participants in nuclear development. The oil crises of the 1970s further increased interest in nuclear power, particularly among countries dependent on imported fossil fuels. France implemented an extensive nuclear construction program that substantially reduced its reliance on fossil fuels for electricity generation.

However, nuclear investment required strong government support and introduced financial risks associated with technological complexity and long construction periods.

Economic Results of Earlier Investments

The economic results of historical nuclear investments have been mixed. Countries that successfully standardized reactor construction and developed specialized industrial supply chains often benefited from reliable, large-scale electricity production. France provides an important example of nuclear energy becoming a major component of national electricity generation. Its nuclear fleet has also supported electricity exports to neighboring countries. However, some projects in Europe and North America experienced significant cost overruns and construction delays.

These difficulties increased borrowing costs and raised questions concerning the competitiveness of new nuclear power plants compared with alternative energy technologies. The International Energy Agency's 2025 assessment concludes that nuclear investment remains heavily influenced by government policy, financing structures, and the management of construction risks.

Transition of Nuclear Plants Toward More Considerate Changes

In response to economic, environmental and safety concerns, nuclear operators have increasingly focused on modernizing existing facilities and developing improved reactor designs. Modernization may involve upgrading cooling systems, improving emergency response procedures, replacing aging components, and installing advanced monitoring technologies.

Some countries have extended the operational lifetimes of existing nuclear facilities after conducting safety assessments and regulatory reviews. At the same time, SMRs and Generation IV reactors have attracted attention because of their potential to provide improved flexibility, safety, and resource efficiency.

These changes reflect a broader transition toward nuclear energy systems designed to respond more effectively to contemporary economic and environmental challenges.

Environmental Background

Detrimental Environmental Implications of Fossil Fuels

Fossil-fuel combustion represents a major contributor to global greenhouse gas emissions. Carbon dioxide released by burning coal, oil, and natural gas accumulates in the atmosphere, intensifying the greenhouse effect and contributing to global warming.

The environmental consequences include rising global temperatures, changing precipitation patterns, sea-level rise, and an increased risk of certain extreme weather events. Fossil-fuel extraction and transportation can also cause environmental damage through habitat destruction, water contamination, and accidental oil spills.

Additionally, air pollutants such as sulfur dioxide, nitrogen oxides, and particulate matter contribute to respiratory illnesses and environmental degradation. These consequences have encouraged governments and international organizations to prioritize cleaner electricity generation.

Initial Environmental Concerns and Movements

Public concern regarding environmental pollution increased significantly during the twentieth century. The publication of Rachel Carson's Silent Spring in 1962 contributed to the growth of modern environmental awareness. Between the 1940s and 1970s, environmental movements increasingly questioned the consequences of industrial development and energy production.

Nuclear power became a controversial subject because of concerns involving radioactive waste, reactor accidents, uranium mining, and nuclear weapons proliferation. The 1979 Three Mile Island accident in the United States intensified public concern and contributed to changes in nuclear

regulation and emergency preparedness. Environmental movements developed differing perspectives on nuclear power. Some opposed its expansion because of radioactive risks, while others increasingly recognized its potential contribution to reducing carbon emissions.

The Chernobyl Accident

On 26 April 1986, a catastrophic nuclear accident occurred at Reactor No. 4 of the Chernobyl Nuclear Power Plant in the Ukrainian Soviet Socialist Republic, then part of the Soviet Union. During an improperly conducted safety test, a combination of reactor design deficiencies and operational failures resulted in a sudden increase in reactor power. Explosions and a subsequent fire released significant quantities of radioactive material into the environment.

The accident caused immediate fatalities among plant personnel and emergency responders, as well as longer-term health and environmental consequences. Large surrounding areas were evacuated, and radioactive contamination affected territories beyond the Soviet Union.

The disaster exposed major weaknesses in nuclear safety procedures, emergency preparedness, and international communication. In response, international organizations strengthened efforts to improve nuclear safety and cooperation. Two important agreements adopted in 1986 established arrangements for notifying other countries about nuclear accidents and requesting international emergency assistance.

Timeline of Events

Date	Description of Event
8th of December 1953	U.S. President Dwight D. Eisenhower delivers the Atoms for Peace speech at the UN General Assembly, promoting the peaceful use of nuclear energy worldwide.
29th of July 1957	The International Atomic Energy Agency (IAEA) is established to promote the safe, secure, and peaceful use of nuclear technology.
26th of April 1986	The Chernobyl Nuclear Power Plant disaster occurs in Ukraine (then USSR), becoming the world's worst nuclear accident and leading to major international nuclear safety reforms.
17th of June 1994	The Convention on Nuclear Safety is adopted to improve nuclear safety standards and international cooperation.

11th of March 2011	The Fukushima Daiichi nuclear accident in Japan follows a massive earthquake and tsunami, prompting many countries to reassess their nuclear energy policies.

Major Actors Involved

France

When looking at how nuclear energy is shaping our world today, a few key players are driving the conversation and the infrastructure. France has long been a leading example of nuclear power, having built up a extensive nuclear fleet back in the 1970s to break free from foreign oil. By 2024, and nuclear energy still generated about two-thirds of the country's electricity. The French are extensive expertise in nuclear engineering and fuel recycling, but they are currently facing financial and logistical challenges associated with aging reactors, soaring maintenance bills, and the ever-present question of what to do with long-term radioactive waste.

China

Meanwhile, China is moving rapidly. As its industrial sector grows, so does its demand for electricity, pushing the country to invest substantially in all kinds of energy projects. Coal is still a major source of electricity in China's power grid, but Beijing is expanding its nuclear capabilities to clean up its air and diversify its options. China is also investing in advanced systems like high-temperature reactors. the IAEA noted that by the end of 2024, nearly half of all new nuclear capacity being built across the globe was in China—a clear sign of what happens when a government commits to long-term industrial planning.

International Atomic Energy Agency (IAEA)

On the regulatory and oversight side, the International Atomic Energy Agency (IAEA) has been an international organisation supporting the safe, secure and peaceful use of nuclear technology since 1957. Operating out of Vienna, its responsibilities include nuclear safeguards, technical cooperation and developing safety standards; safeguards inspectors verify that declared nuclear material is not diverted from peaceful uses. It develops international safety standards, help countries handle radioactive waste, and helps coordinate international responses to nuclear or radiological emergencies. For any country that is new to nuclear energy and lacks its own regulatory framework, the IAEA is an important source of technical assistance.

United Nations Framework Convention on Climate Change (UNFCCC)

Finally, the United Nations Framework Convention on Climate Change (UNFCCC) provides the international framework for climate negotiations. Adopted in 1992, the Convention is supported by a Secretariat that helps organise the annual Conference of the Parties (COP) meetings. Its objective is to stabilise greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous human interference with the climate system.

Through agreements such as the Paris Agreement, parties pursue emissions-reduction goals. These commitments may encourage governments to consider nuclear energy alongside other low-carbon technologies, but they do not require states to build nuclear power plants.

UN Documents and International Agreements

Statute of the International Atomic Energy Agency (1956/1957)

This foundational agreement officially launched the IAEA with a twin mission to accelerate and enlarge the contribution of atomic energy to peace, health, and prosperity worldwide while applying safeguards to verify that nuclear material subject to safeguards is not diverted for military weapons or proliferation purposes. By serving as the legal bedrock for IAEA safeguards, this document establishes the global verification mechanisms that give nations the confidence to trade nuclear technologies peacefully, helping ensure that the expansion of low-carbon power does not compromise international security or fuel geopolitical instability.

Convention on Early Notification of a Nuclear Accident (1986)

Drafted quickly in the direct aftermath of the Chernobyl disaster, this treaty legally binds countries to promptly notify potentially affected states and the IAEA of qualifying nuclear accidents that have resulted or may result in a transboundary radiological release, protecting public health by providing neighboring nations with the rapid, specific data needed to take protective actions. It dismantles state secrecy during technological crises by mandating an open line of communication regarding the time, location, and nature of the release, thereby establishing a framework for timely information-sharing that is vital for maintaining public trust in nuclear operations globally.

Paris Agreement (2015)

This landmark global climate treaty requires parties to prepare, communicate and maintain nationally determined contributions to climate action, which naturally shifts the global energy conversation to consider low-carbon electricity technologies, including nuclear power where appropriate for backing up clean grids and meeting net-zero goals. By encouraging governments to reduce emissions across their economies, this accord provides a powerful political and economic justification for investing in next-generation nuclear systems as a baseline partner to intermittent wind and solar power.

Previous Attempts to Solve the issue

IAEA Safety Guidelines

The IAEA has published detailed, harmonized guidelines that provide requirements and recommendations that countries can incorporate into their national licensing rules, potentially reducing inconsistencies and delays in reactor projects. By creating a common language for risk assessment and safety engineering, these guidelines help countries strengthen domestic regulatory agencies, minimizing the chances of regulatory capture or oversights.

Small Modular Reactor (SMR) Development

The energy sector is exploring smaller units alongside conventional large reactors whose components could be manufactured in factories and transported to installation sites, potentially reducing construction times and some initial financing requirements. This approach could reduce the financial risks of nuclear engineering by allowing utilities to add power incrementally as local demand grows, allowing capital investment to be phased over time; however, actual costs remain uncertain for many designs.

Generation IV International Forum (GIF)

This global cooperative endeavor brings international researchers together to co-develop advanced reactor concepts such as molten salt and fast neutron systems, that aim to improve fuel efficiency, safety and waste management. By pooling the research budgets and laboratory resources of participating countries, the forum supports research into advanced technologies that would be costly and challenging for individual countries or companies to develop alone.

Carbon Pricing and Climate Policies

The implementation of carbon taxes and clean energy credits can place a financial cost on greenhouse gas emissions, potentially improving the competitiveness of nuclear plants that provide reliable, low-carbon electricity. By accounting for some of the environmental costs of fossil-fuel use, these market-based policies can make nuclear energy more competitive despite its high upfront capital costs over the long-term lifespan of the asset.

Possible Solutions

Spent Fuel Recycling and Reprocessing

This advanced technical method extracts leftover uranium and plutonium from used fuel rods to create fresh fuel, potentially reducing the amount of spent fuel requiring disposal, although reprocessing also generates radioactive waste. By using more energy from mined uranium, reprocessing may reduce demand for newly mined uranium but introduces additional costs, waste-management requirements and proliferation risks. Furthermore, some advanced fast-reactor systems could reduce inventories of long-lived actinides, but they do not eliminate the need for safe long-term waste disposal.

Capacity Building and Technology Transfer

To ensure that next-generation systems are deployed safely on a global scale, established nuclear nations could provide developing countries with the rigorous engineering knowledge, regulatory training, and physical safety tools needed to run these advanced facilities. This comprehensive transfer of expertise can reduce gaps in operational safety and oversight, ensuring that as nuclear energy expands into emerging markets to drive economic development, it is managed by a highly skilled, locally empowered workforce. By establishing joint educational institutes, international safety mentorships, and transparent knowledge-sharing agreements, the international community

could help new nuclear adopters avoid the regulatory mistakes of the past, thereby safeguarding the international reputation of next-generation tech.

Public Awareness and Stakeholder Engagement

Building a sustainable nuclear future requires opening direct, honest lines of communication with local communities to address safety concerns head-on, address inaccurate information, and give local voices a real, transparent seat at the planning table. By completely replacing paternalistic "decide-announce-defend" corporate strategies with genuine, early dialogue about environmental risks, economic benefits, and long-term job creation, developers can foster the local trust and reduce the risk of costly political gridlock, legal battles, and project cancellations. True stakeholder engagement means treating host communities as active financial partners in the clean energy transition, offering transparent access to environmental monitoring data and helping local populations benefit from the socioeconomic rewards of the facility.

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Last Remarks

Dear delegates,

I hope the study guide has been of help to all of you to comprehend the genuine impact of nuclear power plants on social, political and mostly environmental aspects of our lives. It is crucial that you all understand the significance of United Nations involvement in these endeavours by adopting international agreements, taking precautionary measures and regulating possible violations.

Thus, I want you all to treat this topic as a race towards a safer future from every perspective and focus on your country's alignment with certain already-existing governing bodies/measures.

Good luck.